

Supplementary Material

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Paper Title: Embedding Neural Scaling Laws into Optimization Dynamics for Large Language Models

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TABLE I. TRAINING CONFIGURATION

Parameter	Value
Framework	PyTorch 2.1
GPU	NVIDIA A100
Precision	FP16 Mixed Precision
Batch Size	64
Sequence Length	1024
Optimizers	Adam, AdamW, LAMB, μ P, SAO
Random Seeds	3
Gradient Clipping	1.0
Dataset	WikiText-103
Model Scales	9.6M, 28.4M, 74.1M, 192.3M
Learning Rate	Fixed across scales
Training Strategy	Mini-batch training

TABLE II. ARCHITECTURAL SPECIFICATIONS AND TRAINABLE PARAMETER COUNTS OF EVALUATED MODELS

Model	Layers (L)	d_{model}	Heads (h)	FFN Dim	Approx. Parameters (N)
SIO-S	6	256	4	1024	9.6M
SIO-M	12	512	8	2048	48.2M
SIO-L	18	768	12	3072	108.5M
SIO-XL	24	1024	16	4096	192.3M

TABLE III. QUANTITATIVE EVALUATION OF SCALE-INVARIANT OPTIMIZATION ACROSS MODEL SCALES

Model	Optimizer	Final Loss $\downarrow$	Convergence Steps $\downarrow$	$E_{\text{scale}} \downarrow$	$S_{\text{inv}} \downarrow$	$G_{\text{CS}} \downarrow$	$\delta_L \downarrow$
SAO-S	Adam	2.91 ± 0.03	$9,800 \pm 110$	0.084	0.021	0.017	0.043 ± 0.003
SAO-S	AdamW	2.85 ± 0.03	$9,200 \pm 100$	0.079	0.019	0.015	0.041 ± 0.003
SAO-S	LAMB	2.82 ± 0.02	$8,400 \pm 95$	0.068	0.016	0.013	0.034 ± 0.002
SAO-S	μ P	2.80 ± 0.02	$8,100 \pm 90$	0.061	0.014	0.011	0.031 ± 0.002
SAO-S	SAO	2.78 ± 0.01	$7,600 \pm 80$	0.052	0.011	0.008	0.026 ± 0.002
SAO-M	Adam	2.64 ± 0.04	$11,200 \pm 140$	0.097	0.028	0.022	0.051 ± 0.004
SAO-M	AdamW	2.58 ± 0.04	$10,600 \pm 130$	0.091	0.024	0.019	0.048 ± 0.003
SAO-M	LAMB	2.53 ± 0.03	$9,800 \pm 115$	0.078	0.020	0.017	0.041 ± 0.003
SAO-M	μ P	2.51 ± 0.03	$9,400 \pm 105$	0.069	0.018	0.014	0.038 ± 0.003
SAO-M	SAO	2.49 ± 0.02	$8,900 \pm 95$	0.060	0.013	0.010	0.031 ± 0.002
SAO-L	Adam	2.48 ± 0.05	$13,500 \pm 170$	0.121	0.034	0.029	0.067 ± 0.005
SAO-L	AdamW	2.43 ± 0.05	$12,800 \pm 160$	0.115	0.031	0.025	0.063 ± 0.004
SAO-L	LAMB	2.39 ± 0.04	$11,700 \pm 145$	0.096	0.026	0.022	0.054 ± 0.004
SAO-L	μ P	2.37 ± 0.03	$11,200 \pm 135$	0.085	0.023	0.019	0.049 ± 0.003
SAO-L	SAO	2.36 ± 0.02	$10,400 \pm 120$	0.071	0.016	0.012	0.038 ± 0.003

TABLE IV. ABLATION ANALYSIS OF SCALE-AWARE COMPONENTS IN SAO

Variant	Final Loss $\downarrow$	$E_{\text{scale}} \downarrow$	$S_{\text{inv}} \downarrow$	$G_{\text{CS}} \downarrow$	$\delta_L \downarrow$
Full SAO	2.36	0.071	0.016	0.012	0.038
Without Normalized Scaling Exponents	2.45	0.095	0.029	0.021	0.056
Without Scale-Aware Update Scaling	2.50	0.116	0.034	0.028	0.073

TABLE V. COMPUTATIONAL OVERHEAD COMPARISON OF OPTIMIZERS

Optimizer	Relative Training Overhead (vs. Adam)
Adam	1.00×
AdamW	1.01×
LAMB	1.03×
μ P	1.02×
SAO	1.04×

Note: Relative overhead values are normalized with respect to Adam-based optimization under identical training configurations.